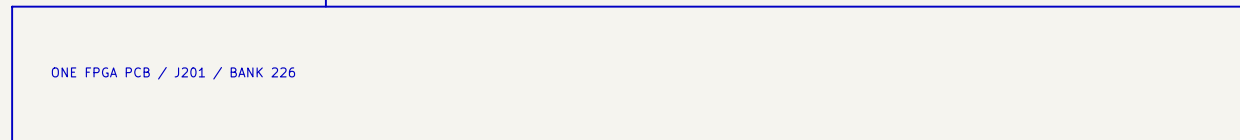
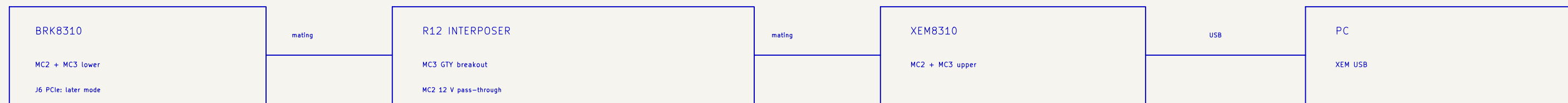


R12 SINGLE-LINK STUDY | ONE COMPLETE 19-CONTACT CUSTOM FPGA LINK

Single custom FPGA link with data, translated JTAG, shared return and protected 12 V; other MC3 banks pass through.



02 / PORT 1 - MC3 BANK 226 - TYPE-D J201

03 / MC3 BANK 225 - COMPLETE PASS-THROUGH

04 / MC3 BANK 224 - COMPLETE PASS-THROUGH

05 / MC2 POWER + MC3 BANK-67 GPIO

06 / MC1 1-40 PASS-THROUGH

07 / MC1 41-80 PASS-THROUGH

08 / MC2 21-80 PASS-THROUGH

SIGNAL CHOICE

Each link uses two recording pairs, one control pair and a Reserved pair. Bank 226 serves the single J201 link

MC1/MC2 all 80 pins each pass through; selected Bank-226 data and five Bank-67 GPIO isolate lower BRK contacts

Concrete JTAG and power circuits are on sheets 09–10. Load limits and fabrication release remain unqualified.

Custom Type-D carries protected 12 V on contact 19. Dedicated custom cable only. NOT FOR FABRICATION

09 / PORT 1 JTAG BRIDGE + PROBE

10 / SINGLE-LINK PROTECTED 12 V BRANCH

H1

M3 / 3.2 mm



H2

M3 / 3.2 mm



H3

M3 / 3.2 mm



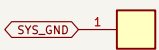
H4

M3 / 3.2 mm



#FLG1

EXTERNAL SOURCE / SYS_GND



#FLG2

EXTERNAL_SOURCE / VDCIN_EXTERNAL_12V_CANDIDATE



#FLG3

EXTERNAL SOURCE / XEM_1V2_DDR



#FLG4

EXTERNAL SOURCE / P1_VTREF_1VB_TARGET

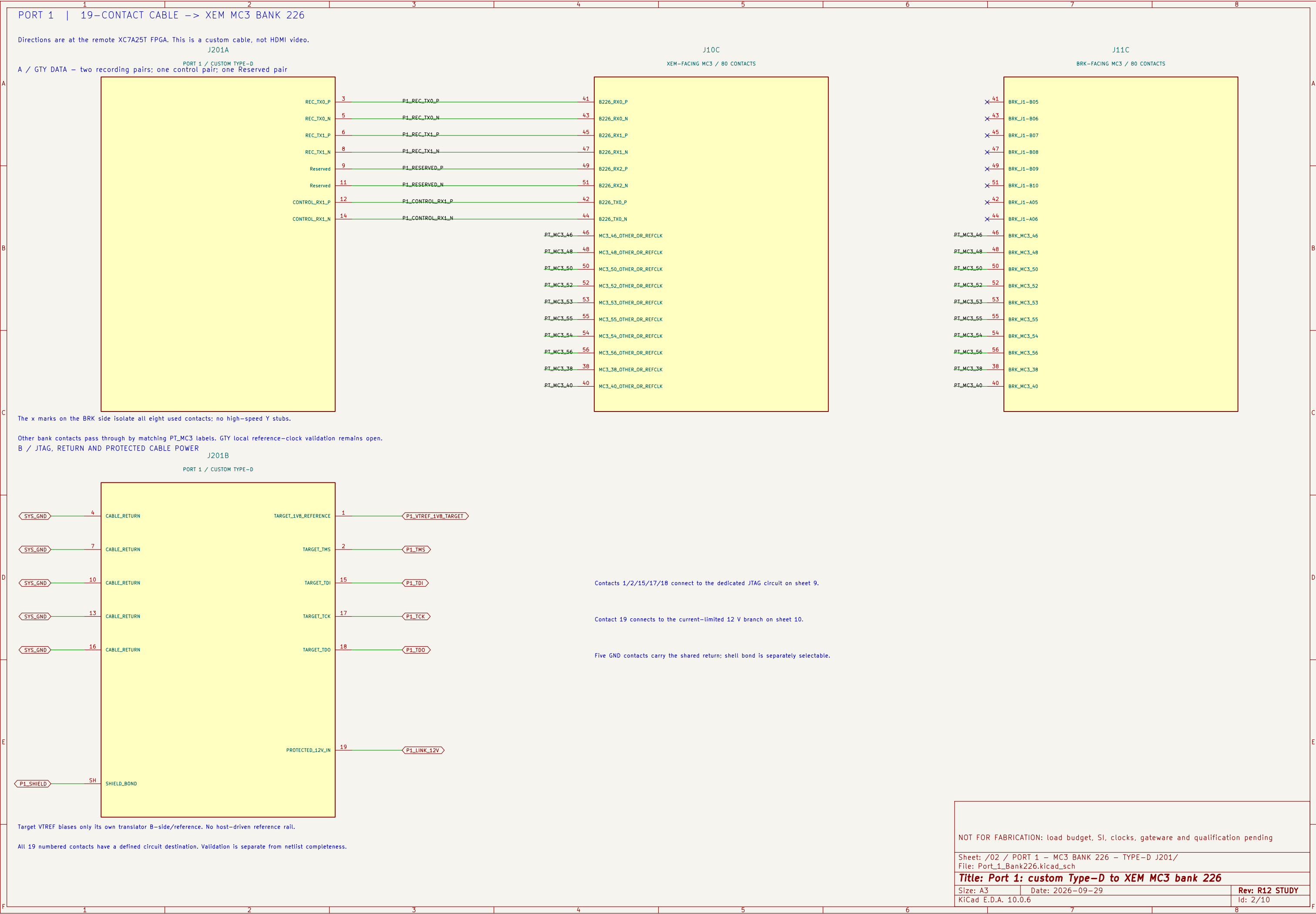


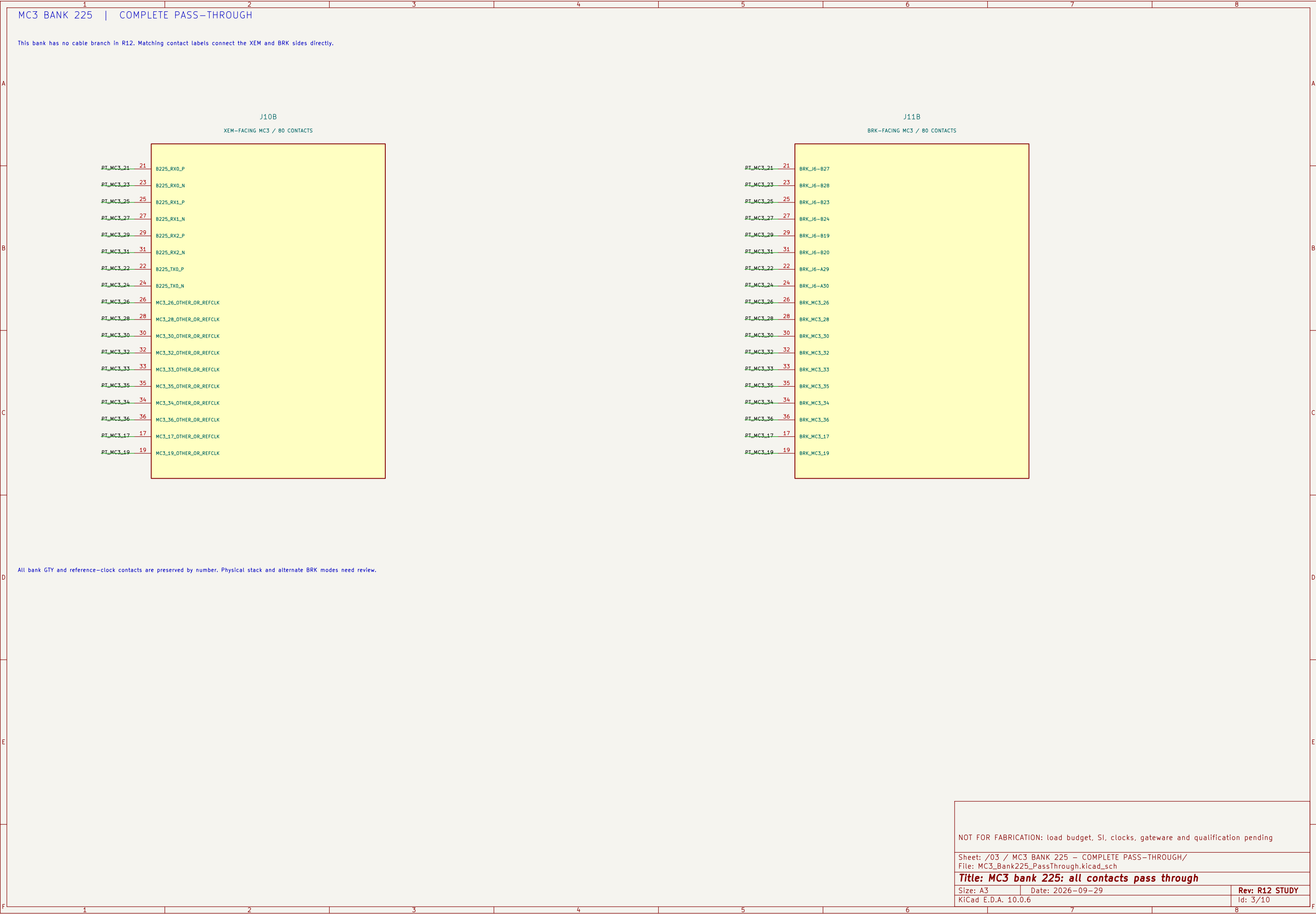
NOT FOR FABRICATION: load budget, SI, clocks, gateware and qualification pending

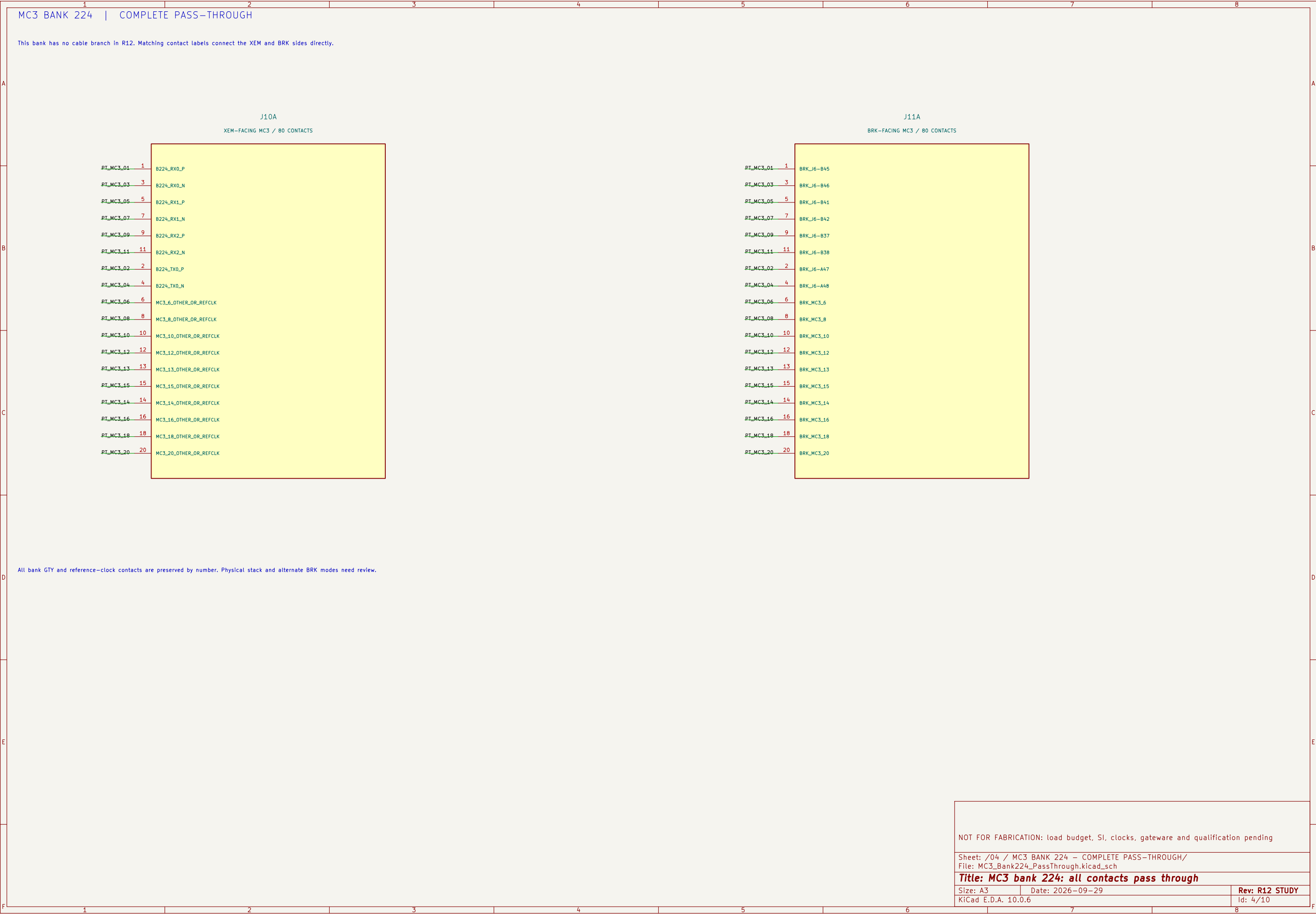
Sheet: /
File: BRK8310_Single_Link_Interposer_R12.kicad_sch

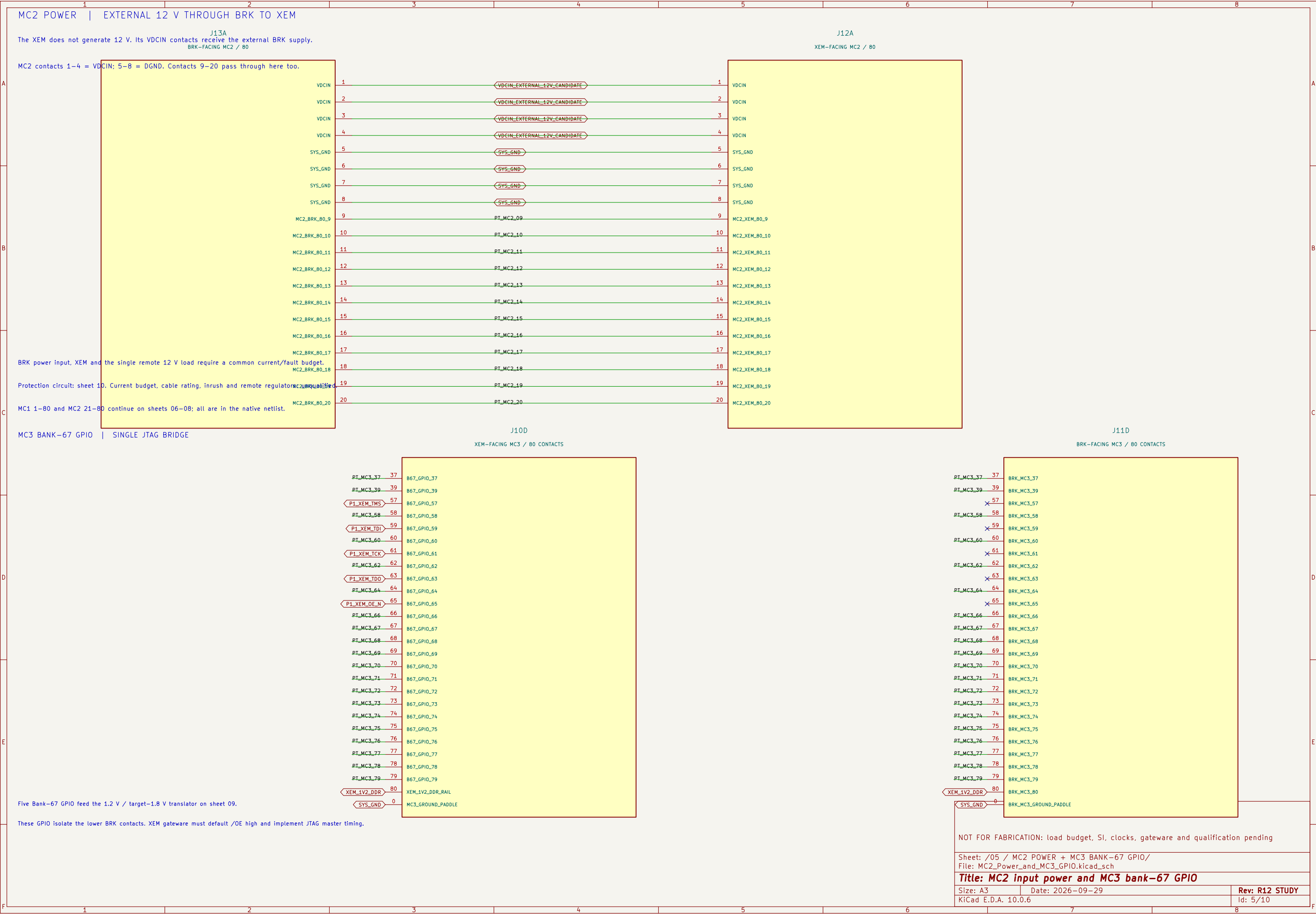
Title: XEM8310 / BRK8310 single-link interposer - circuit map

Size: A2	Date: 2026-09-29	Rev: R12 STUDY
KiCad E.D.A. 10.0.6		Id: 1/10









Five Bank-67 GPIO feed the 1.2 V / target-1.8 V translator on sheet 09.

These GPIO isolate the lower BRK contacts. XEM gateware must default /OE high and implement JTAG master timing.

NOT FOR FABRICATION: load budget, SI, clocks, gateware and qualification pending

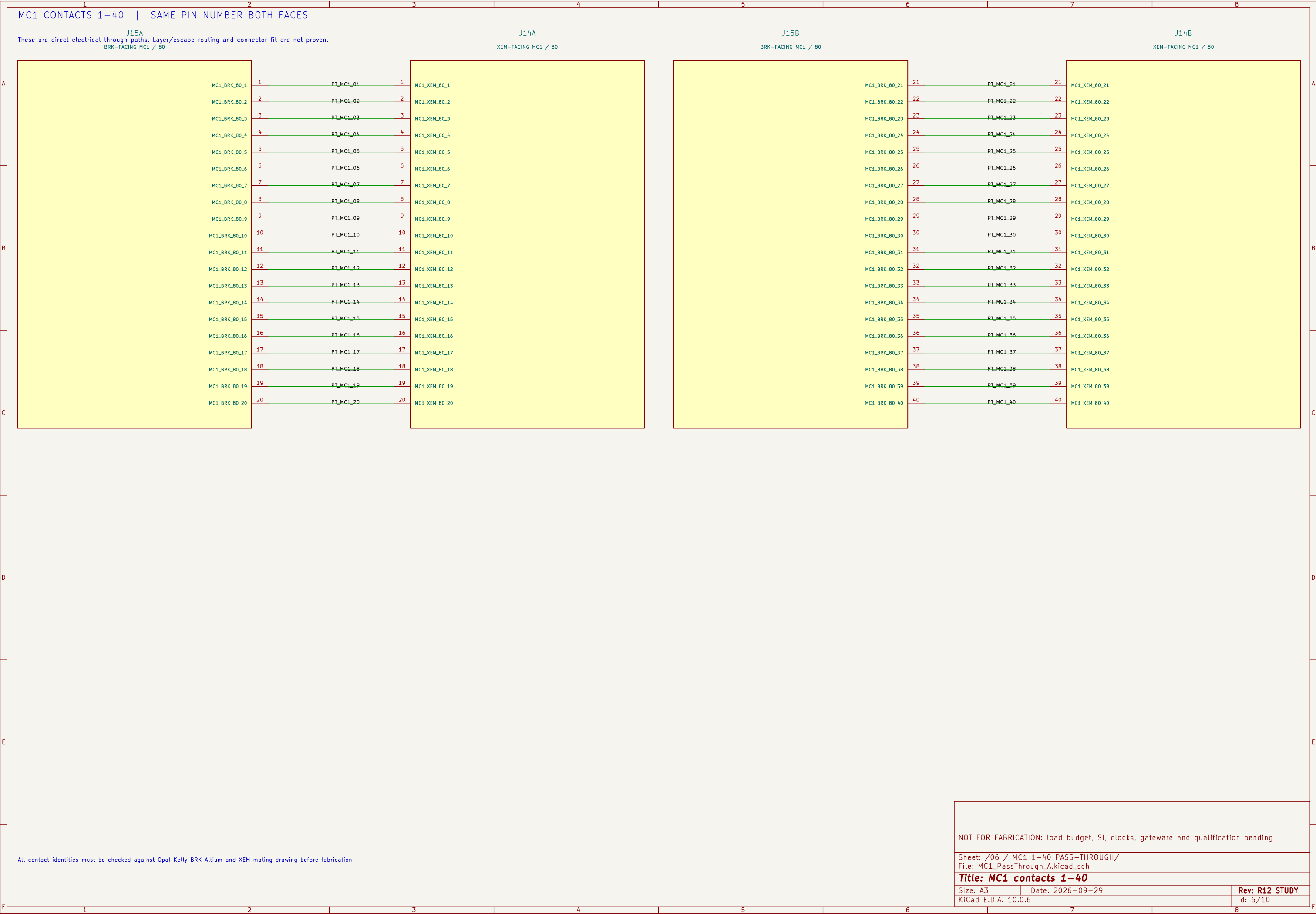
Sheet: /05 / MC2 POWER + MC3 BANK-67 GPIO/
File: MC2_Power_and_MC3_GPIO.kicad_sch

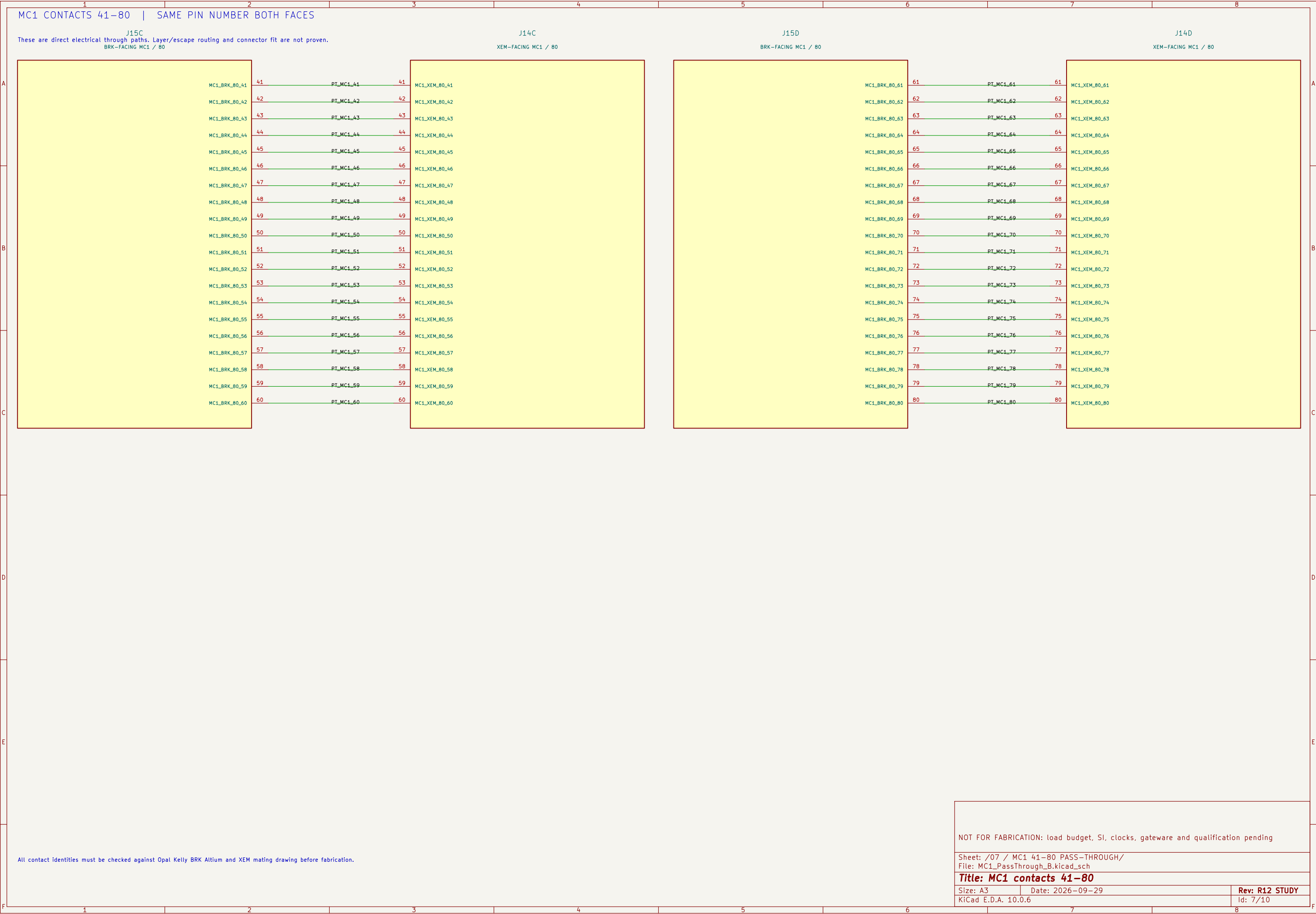
Title: MC2 input power and MC3 bank-67 GPIO

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-29

Rev: R12 STUDY
Id: 5/10





MC2 CONTACTS 21–80 | SAME PIN NUMBER BOTH FACES

J13B
These are direct electrical through paths. Layer/escape routing and connector fit are not proven.
BRK-FACING MC2 / 80

MC2_BRK_80_21	21	PT_MC2_21	21	MC2_XEM_80_21
MC2_BRK_80_22	22	PT_MC2_22	22	MC2_XEM_80_22
MC2_BRK_80_23	23	PT_MC2_23	23	MC2_XEM_80_23
MC2_BRK_80_24	24	PT_MC2_24	24	MC2_XEM_80_24
MC2_BRK_80_25	25	PT_MC2_25	25	MC2_XEM_80_25
MC2_BRK_80_26	26	PT_MC2_26	26	MC2_XEM_80_26
MC2_BRK_80_27	27	PT_MC2_27	27	MC2_XEM_80_27
MC2_BRK_80_28	28	PT_MC2_28	28	MC2_XEM_80_28
MC2_BRK_80_29	29	PT_MC2_29	29	MC2_XEM_80_29
MC2_BRK_80_30	30	PT_MC2_30	30	MC2_XEM_80_30
MC2_BRK_80_31	31	PT_MC2_31	31	MC2_XEM_80_31
MC2_BRK_80_32	32	PT_MC2_32	32	MC2_XEM_80_32
MC2_BRK_80_33	33	PT_MC2_33	33	MC2_XEM_80_33
MC2_BRK_80_34	34	PT_MC2_34	34	MC2_XEM_80_34
MC2_BRK_80_35	35	PT_MC2_35	35	MC2_XEM_80_35
MC2_BRK_80_36	36	PT_MC2_36	36	MC2_XEM_80_36
MC2_BRK_80_37	37	PT_MC2_37	37	MC2_XEM_80_37
MC2_BRK_80_38	38	PT_MC2_38	38	MC2_XEM_80_38
MC2_BRK_80_39	39	PT_MC2_39	39	MC2_XEM_80_39
MC2_BRK_80_40	40	PT_MC2_40	40	MC2_XEM_80_40

J12B
XEM-FACING MC2 / 80

J13C
BRK-FACING MC2 / 80

MC2_BRK_80_41	41	PT_MC2_41	41	MC2_XEM_80_41
MC2_BRK_80_42	42	PT_MC2_42	42	MC2_XEM_80_42
MC2_BRK_80_43	43	PT_MC2_43	43	MC2_XEM_80_43
MC2_BRK_80_44	44	PT_MC2_44	44	MC2_XEM_80_44
MC2_BRK_80_45	45	PT_MC2_45	45	MC2_XEM_80_45
MC2_BRK_80_46	46	PT_MC2_46	46	MC2_XEM_80_46
MC2_BRK_80_47	47	PT_MC2_47	47	MC2_XEM_80_47
MC2_BRK_80_48	48	PT_MC2_48	48	MC2_XEM_80_48
MC2_BRK_80_49	49	PT_MC2_49	49	MC2_XEM_80_49
MC2_BRK_80_50	50	PT_MC2_50	50	MC2_XEM_80_50
MC2_BRK_80_51	51	PT_MC2_51	51	MC2_XEM_80_51
MC2_BRK_80_52	52	PT_MC2_52	52	MC2_XEM_80_52
MC2_BRK_80_53	53	PT_MC2_53	53	MC2_XEM_80_53
MC2_BRK_80_54	54	PT_MC2_54	54	MC2_XEM_80_54
MC2_BRK_80_55	55	PT_MC2_55	55	MC2_XEM_80_55
MC2_BRK_80_56	56	PT_MC2_56	56	MC2_XEM_80_56
MC2_BRK_80_57	57	PT_MC2_57	57	MC2_XEM_80_57
MC2_BRK_80_58	58	PT_MC2_58	58	MC2_XEM_80_58
MC2_BRK_80_59	59	PT_MC2_59	59	MC2_XEM_80_59
MC2_BRK_80_60	60	PT_MC2_60	60	MC2_XEM_80_60

J12C
XEM-FACING MC2 / 80

J13D
BRK-FACING MC2 / 80

MC2_BRK_80_61	61	PT_MC2_61	61	MC2_XEM_80_61
MC2_BRK_80_62	62	PT_MC2_62	62	MC2_XEM_80_62
MC2_BRK_80_63	63	PT_MC2_63	63	MC2_XEM_80_63
MC2_BRK_80_64	64	PT_MC2_64	64	MC2_XEM_80_64
MC2_BRK_80_65	65	PT_MC2_65	65	MC2_XEM_80_65
MC2_BRK_80_66	66	PT_MC2_66	66	MC2_XEM_80_66
MC2_BRK_80_67	67	PT_MC2_67	67	MC2_XEM_80_67
MC2_BRK_80_68	68	PT_MC2_68	68	MC2_XEM_80_68
MC2_BRK_80_69	69	PT_MC2_69	69	MC2_XEM_80_69
MC2_BRK_80_70	70	PT_MC2_70	70	MC2_XEM_80_70
MC2_BRK_80_71	71	PT_MC2_71	71	MC2_XEM_80_71
MC2_BRK_80_72	72	PT_MC2_72	72	MC2_XEM_80_72
MC2_BRK_80_73	73	PT_MC2_73	73	MC2_XEM_80_73
MC2_BRK_80_74	74	PT_MC2_74	74	MC2_XEM_80_74
MC2_BRK_80_75	75	PT_MC2_75	75	MC2_XEM_80_75
MC2_BRK_80_76	76	PT_MC2_76	76	MC2_XEM_80_76
MC2_BRK_80_77	77	PT_MC2_77	77	MC2_XEM_80_77
MC2_BRK_80_78	78	PT_MC2_78	78	MC2_XEM_80_78
MC2_BRK_80_79	79	PT_MC2_79	79	MC2_XEM_80_79
MC2_BRK_80_80	80	PT_MC2_80	80	MC2_XEM_80_80

J12D
XEM-FACING MC2 / 80

All contact identities must be checked against Opal Kelly BRK Altium and XEM mating drawing before fabrication.

NOT FOR FABRICATION: load budget, SI, clocks, gateway and qualification pending
Sheet: /08 / MC2 21–80 PASS-THROUGH/
File: MC2_PassThrough.kicad_sch
Title: MC2 contacts 21–80
Size: A3Date: 2026–09–29Rev: R12 STUDY
KiCad E.D.A. 10.0.6Id: 8/10

All contact identities must be checked against Opal Kelly BRK Altium and XEM mating drawing before fabrication.

NOT FOR FABRICATION: load budget, SI, clocks, gateware and qualification pending

Sheet: /08 / MC2 21-80 PASS-THROUGH/
File: MC2_PassThrough.kicad_sch

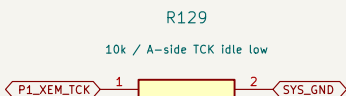
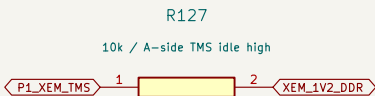
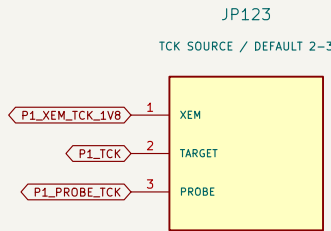
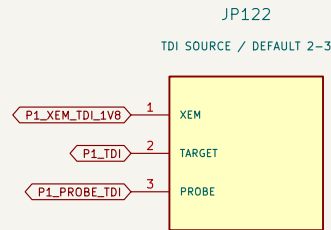
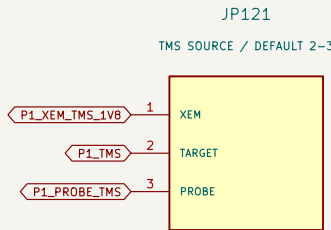
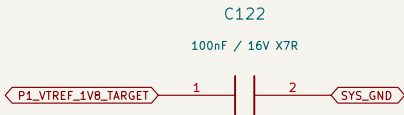
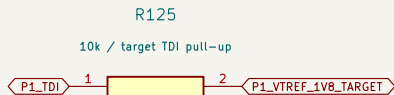
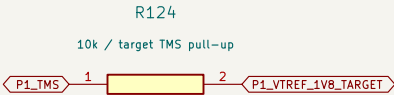
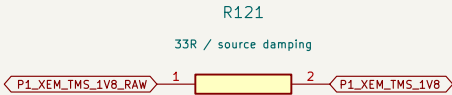
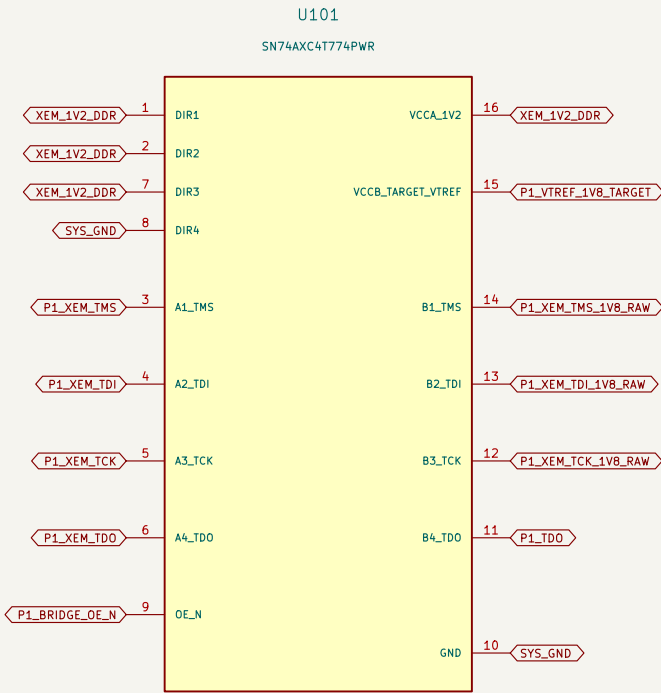
Title: MC2 contacts 21-80

Size: A3	Date: 2026-09-29	Rev: R12 STUDY
KiCad E.D.A. 10.0.6		Id: 8/10

PORT 1 | INDEPENDENT 1.2 V / TARGET 1.8 V JTAG

Target-derived VTREF supplies the translator B rail; the host never drives VTREF. The host does not supply the target reference.

Default: all three source shunts 2–3 PROBE; ARM jumper open. Move all three to 1–2 for XEM master operation.



NOT FOR FABRICATION: load budget, SI, clocks, gateware and qualification pending

Sheet: /09 / PORT 1 JTAG BRIDGE + PROBE/
File: Port_1_JTAG.kicad_sch

Title: Port 1: XEM JTAG bridge and recovery source selection

Size: A3	Date: 2026-09-29	Rev: R12 STUDY
KiCad E.D.A. 10.0.6		Id: 9/10

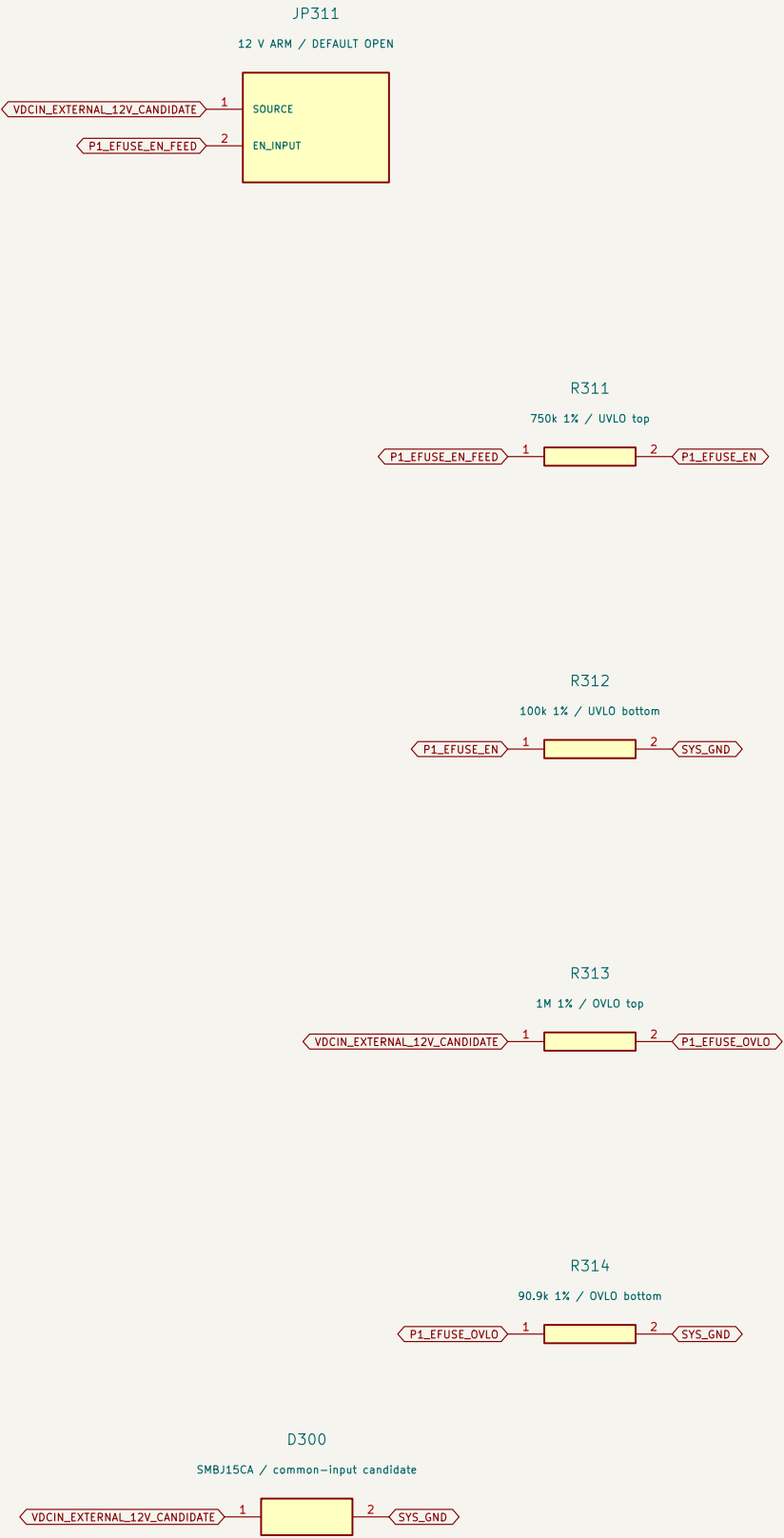
DIR1/2/3 = HIGH: TMS/TDI/TCK to target. DIR4 = LOW: target TDO to XEM. /OE defaults HIGH.

MC3.80 is +1.2V_DDR (250 mA shared user budget). Target pin 1 bias/inrush load, signal timing and gateware require review.

PORT 1 | PROTECTED 12 V TO CONTACT 19

A latch-off eFuse protects the single cable. Enable shunt is open by default; install only after the remote load is reviewed.

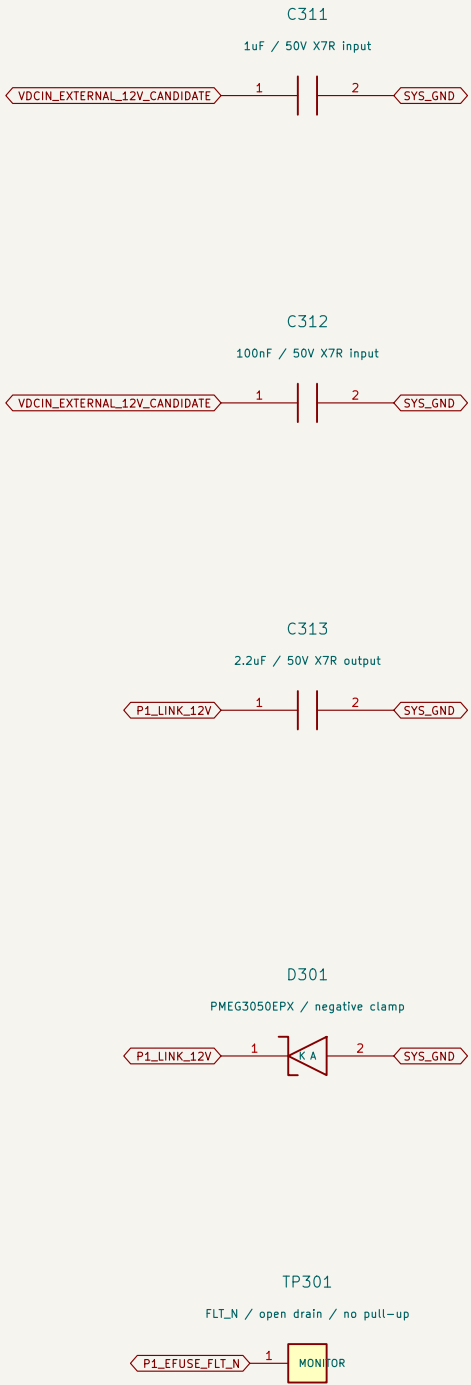
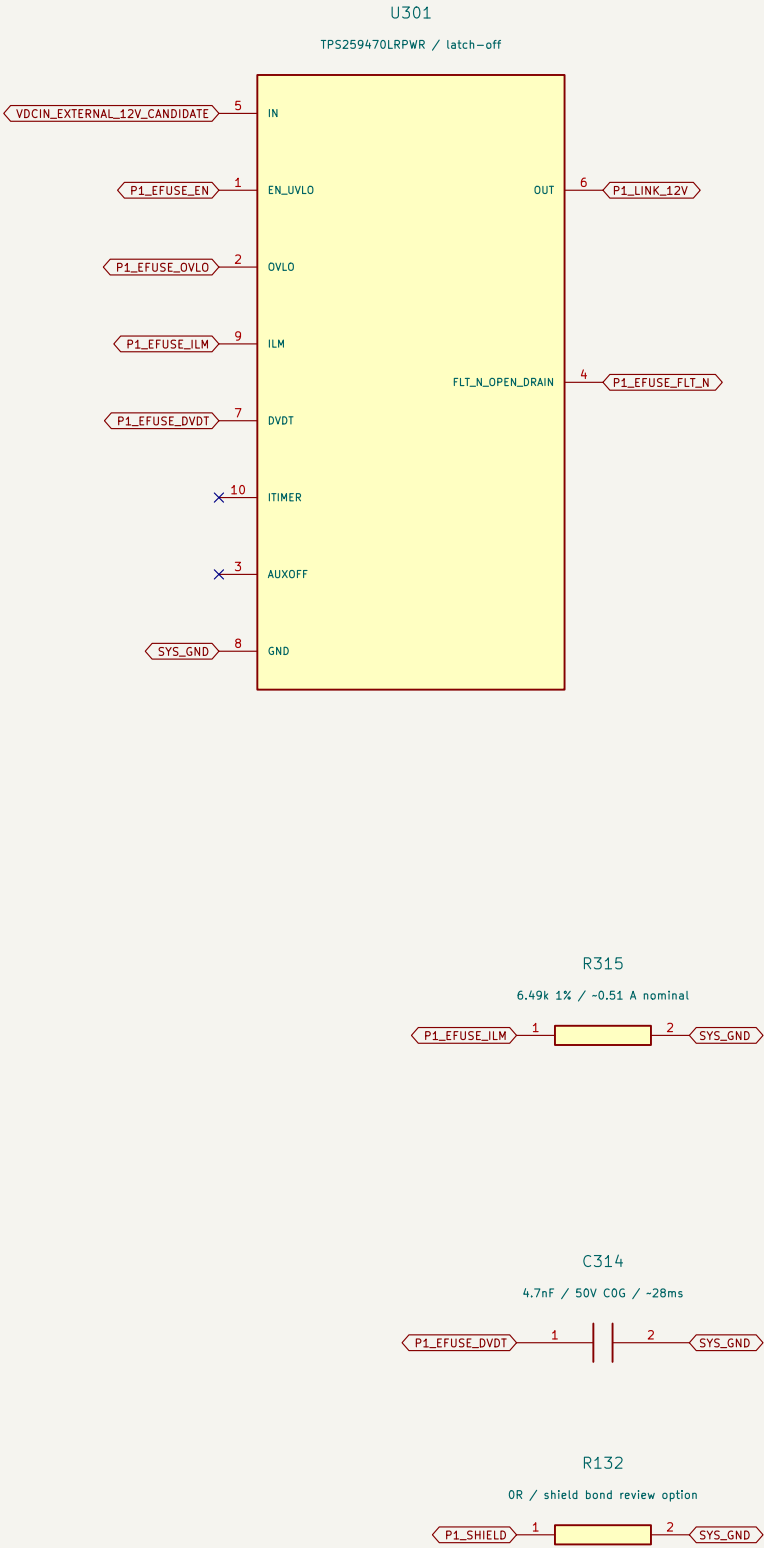
Approx. 0.51 A nominal is an adjustable study setting, not a qualified cable limit or a proven FPGA/ASIC power budget.



Five cable GND contacts carry return current. Shell bonds through OR; shell is not the only load return.

UVLO approx.10.2 V; OVLO approx.14.4 V. ITIMER/AUXOFF NC; FLT_N has no host-voltage pull-up.

Use a clean regulated 12 V source. Shared SMBJ15CA does not limit the XEM input to its 15 V operating maximum.



NOT FOR FABRICATION: load budget, SI, clocks, gateware and qualification pending

Sheet: /10 / SINGLE-LINK PROTECTED 12 V BRANCH/
File: Port_1_Power.kicad_sch

Title: Port 1: protected external 12 V branch and shield bond

Size: A3	Date: 2026-09-29	Rev: R12 STUDY
KiCad E.D.A. 10.0.6		Id: 10/10